ANANYA AGARWAL

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SUMMARY

Information Technology undergraduate at VIT (2027) with hands-on software development experience at Reliance Jio. Skilled in Python, FastAPI, React, REST APIs, PostgreSQL, and Docker, with experience building backend services, real-time dashboards, and automated tests. Strong foundation in Data Structures & Algorithms, OOP, DBMS, and Operating Systems.

EDUCATION

Vellore Institute of Technology,Vellore – B.Tech, Information Technology | **CGPA: 8.79** Aug 2023 – May 2027
Ryan International School (CBSE) — Class XII: 92.2% (2022) | Class X: 93.2% (2020)

EXPERIENCE

Reliance Jio — Software Developer Intern May 2026 – June 2026

- Contributed to **Telecom360**, a large-scale multi-operator telecom lifecycle and operations intelligence platform, working across **7 full-stack modules** spanning **SIM inventory & lifecycle management**, **KYC workflows**, multi-stage activation orchestration, MNP, and usage/recharge services.
- Developed backend **REST APIs** with **Python, FastAPI, SQLAlchemy, Pydantic** with **database modelling** on **PostgreSQL**; implemented **concurrency-safe number reservation** using **Redis**-based distributed locking.
- Implemented **JWT/RBAC** security and **WebSocket**-driven real-time, role-based operational dashboards using **React 19**, Vite, Tailwind CSS, React Router, Recharts, and React Flow.
- Wrote unit and integration tests with **PyTest** and **Vitest**. Contributed to production-ready features currently **undergoing integration, validation, and pre-deployment testing as part of enterprise rollout**.

PROJECTS

Docker-Based Systems & Security Engineering *FastAPI, Kafka, Spark, Cassandra*

- Built two containerized systems spanning **9+ services**: an ED congestion forecasting platform (**FastAPI, Kafka, Spark, Cassandra, Elasticsearch/Kibana**) and a container security framework (**Docker Bench, Falco, Wazuh**) covering streaming analytics, **observability**, threat detection, and automated response.

AI Image Classifier – Convolutional Neural Network (CNN) *Python, Machine Learning, Deep Learning*

- Built an AI model that automatically recognizes and classifies images using a **LeNet Convolutional Neural Network (CNN)** in Python; accelerated deep learning model training by ~**35%** via multithreaded parallel processing.

Quiz Platform – AI-Automated Test Case Design & Evaluation *Web App, AI-Assisted Testing*

- Built a **free quiz web platform** end-to-end, then used **AI coding tools** to generate and execute automated test suites covering **edge, boundary, and stress cases** — reviewing and refining every **AI-generated test** against requirements before release.

Hospital Management System *Node.js, Express.js, MongoDB, Bootstrap*

- Developed a full-stack application with **role-based authentication**, **RESTful API design and integration**, and a responsive dashboard-driven interface backed by optimized **MongoDB** queries.

GeoRift – Hackathon (2nd of 450 teams) *React.js, Three.js, Python, TensorFlow, OpenCV*

- Built an interactive AI-powered platform to simulate autonomous deep-sea exploration using computer vision and intelligent mission planning, delivering real-time visualization and securing **2nd place among 450 teams**.

PATENT

Published Patent – Topology-Aware Distributed Cross-Device Governance System *Federated Learning*

- Distributed screen-time governance system using **federated learning** and topology-aware device coordination.

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, multithreading, message queues
Languages: Python, C, C++, JavaScript, SQL, HTML, CSS
Frameworks: React, FastAPI, Node.js, Express.js, SQLAlchemy, Tailwind CSS
Databases: PostgreSQL, MongoDB, Redis, Cassandra (SQL & NoSQL design, query optimization)
Testing: PyTest, Vitest, unit testing, test case design, test-driven development
AI Coding Tools: GitHub Copilot, Claude, Cursor, ChatGPT — AI-assisted development and code review
Cloud & DevOps: AWS, Docker, Docker Compose, CI/CD fundamentals, Kafka, Git, GitHub, WebSockets, JWT/RBAC

LEADERSHIP

- **PR & Marketing Head, CSED VIT** — led a 10+ member team and executed events with **400+ participants**.
- **Sponsorship Coordinator, Riviera VIT** — engaged 15+ potential sponsors and coordinated sponsorship deliverables.